

Population structure

Biologia della Conservazione 25/26

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- ▶ Individuals rarely mate completely at random!
- ▶ Even within species, there's often geographically-restricted mating among individuals
- ▶ Individuals tend to mate with individuals from the same, or closely related sets of populations
- ▶ This form of non-random mating is called population structure and can have profound effects on the distribution of genetic variation within and among natural populations
- ▶ In most species individuals are scattered in local conglomerates that are called *demes* (*i.e.*, local populations or subpopulations)

- ▶ Summarizing population structure is often a key initial stage in population genetic and genomic analyses
- ▶ We can define inbreeding as having parents that are more closely related to each other than two individuals drawn at random from some reference population
- ▶ The question that naturally arises is: Which reference population should we use? While I might not look inbred in comparison to allele frequencies in Italy, where I am from, my parents certainly are not two individuals drawn at random from the world-wide population

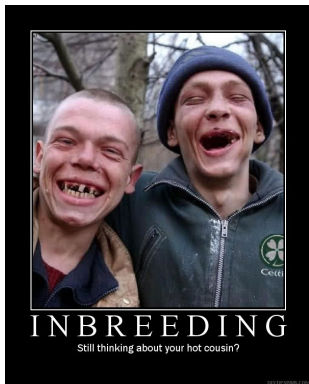


Figure 1: Inbreeding couple



Figure 2: Inbreeding couple

► There cannot be a class on inbreeding without these pictures!!

- ▶ In its most simple definition inbreeding is the mating between related individuals. An individual is considered inbred if its parents share a common ancestor
- ▶ In large population inbreeding can happen because of the non-random assortment of gametes
- ▶ In many tree species spatially closer trees are more likely to be related than trees further apart

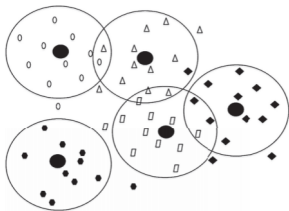


Figure 3: Image after Allendorf et al. (2022), pg. 178

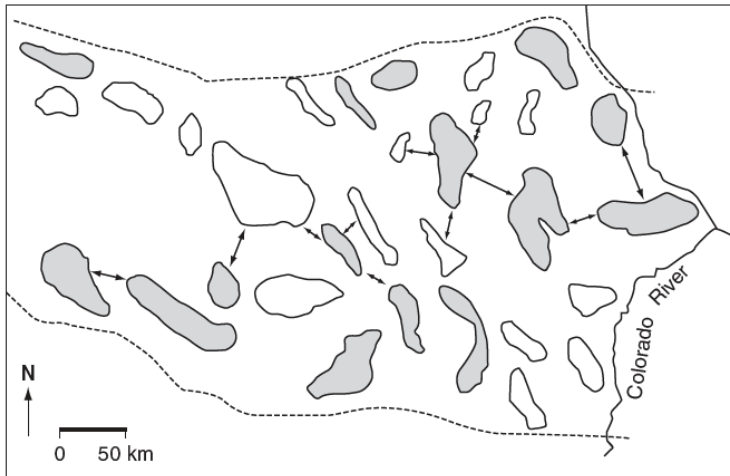


Figure 4: After Akçakaya, Mills, & Doncaster (2007)

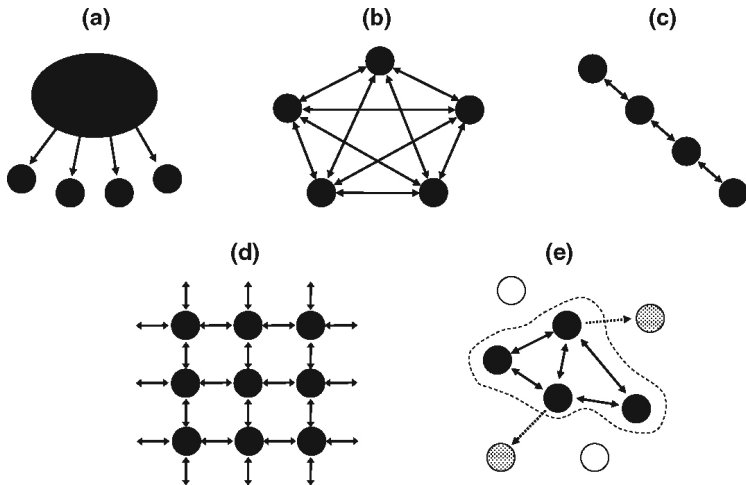


Figure 5: After Frankham, Ballou, & Briscoe (2010)

- ▶ A metapopulation is defined as a set of discrete populations (demes) of the same species, in the same general geographical area, potentially able to exchange individuals through migration, dispersal, or human-mediated movement (Akçakaya et al., 2007; Hanski & Simberloff, 1997)
- ▶ This subdivision (or structure) of populations can be natural and/or human driven
- ▶ Regardless of the cause, the existence of population structure in many species forces us to think at the genetic variation within a species at two primary levels:
 1. Genetic variation within local demes
 2. Genetic variation between local demes

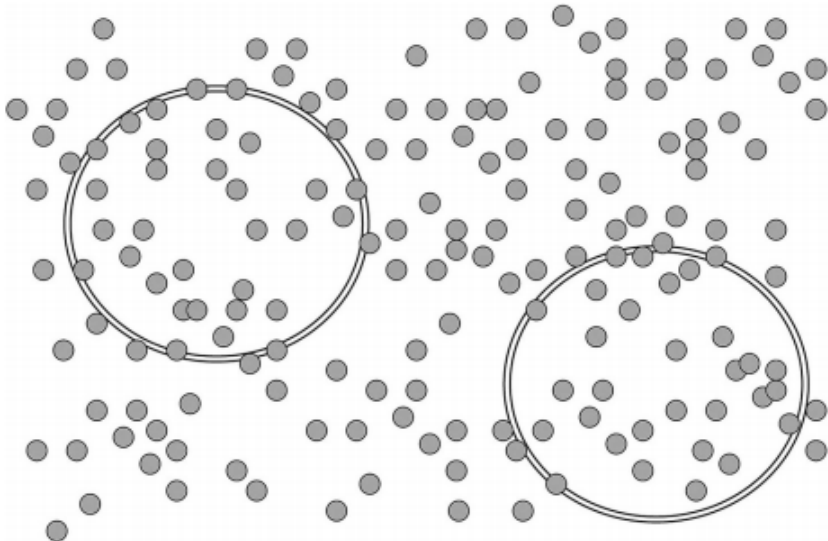


Figure 6: After Allendorf et al. (2022), p. 184

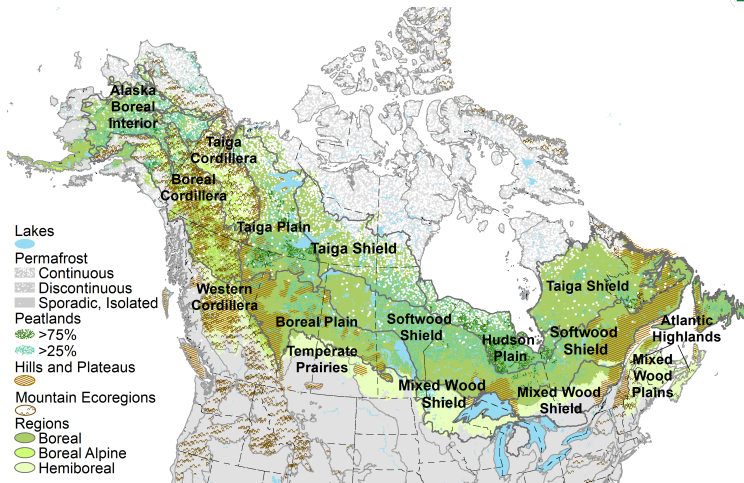


Figure 7: Boreal forest distribution in North America. Available [here](#)

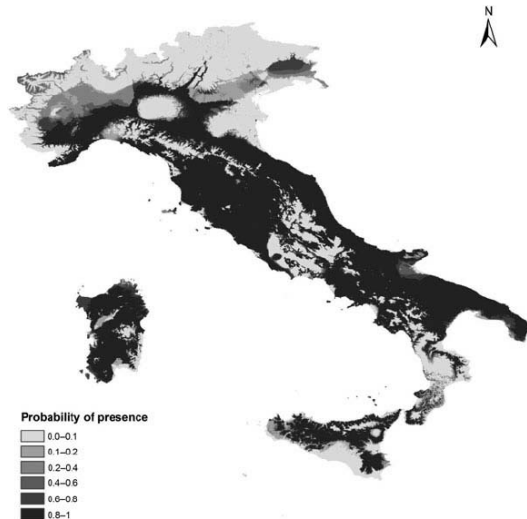


Figure 8: Maxent prediction of *Hystrix cristata* distribution After Mori, Sforzi, & Di Febbraro (2013)

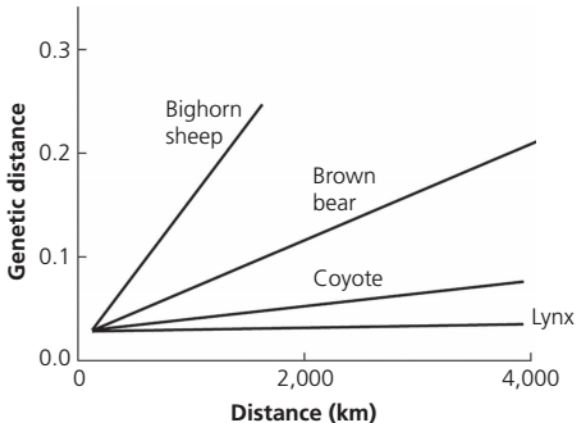


Figure 9: Individuals that are very distant from each other may present significant differences at the genetic level: isolation by distance. Modified from Forbes & Hogg (1999) with unpublished data from M.K. Schwartz (Allendorf et al., 2022).

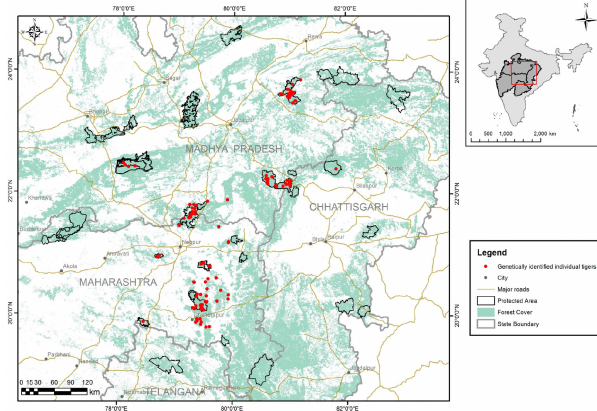


Figure 10: Describing the patterns and amount of differentiation among demes is of the uttermost importance to develop effective conservation strategies. Study area in Thatte, Joshi, Vaidyanathan, Landguth, & Ramakrishnan (2018). Genetically identified tigers as red dots

- ▶ Frankham et al. (2011) describe the risks of outbreeding depression when translocating individuals from genetically distinct populations to supplement imperiled ones

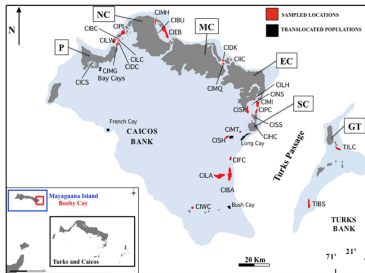


Figure 11: After Welch et al. (2017).

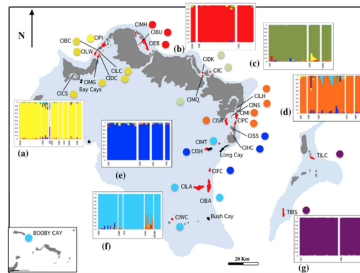


Figure 12: After Welch et al. (2017).

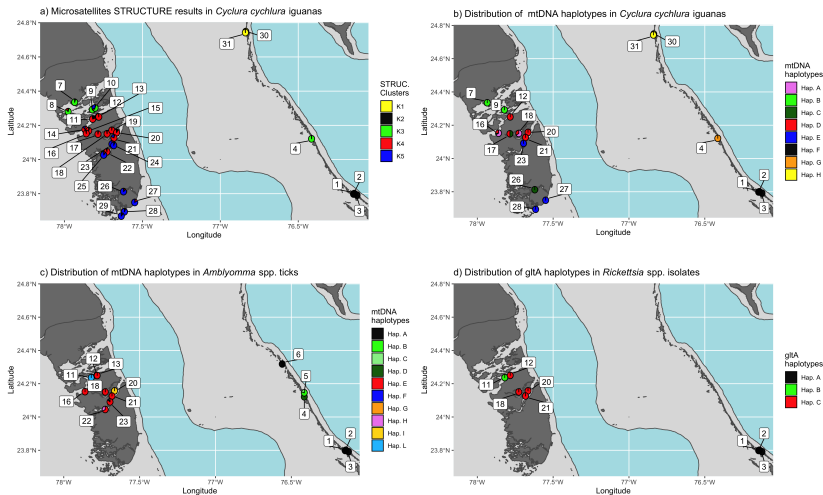


Figure 13: Patterns of differentiation in *Cyclura cyclura* populations. After Colosimo et al. (2021)

- ▶ Developing priorities for the conservation of a species often requires an understanding of adaptive genetic differentiation among populations (Allendorf et al., 2022)
- ▶ To understand how genetic variation is distributed at different loci within subdivided populations it is necessary to look at different interacting forces like gene flow, genetic drift and natural selection
- ▶ F statistics (Malécot, 1948; Wright, 1931, 1951) are undoubtedly the most used indexes to describe genetic differentiation in and between demes
- ▶ F statistics are, in short, inbreeding coefficients measuring different types of inbreeding at different levels

- ▶ We recognize three primary coefficients: F_{IS} , F_{IT} , and F_{ST}
- ▶ F_{IS} measures Hardy-Weinberg departures within local demes
- ▶ F_{ST} measures divergence in allele frequencies among demes
- ▶ F_{IT} measures Hardy-Weinberg departures in the metapopulation due to F_{IS} (nonrandom mating within local demes) and F_{ST} (allele frequency divergence among demes)
- ▶ The original F statistics developed by Wright focused on loci with only two alleles
- ▶ Nei (1977) extended these measures to multiple alleles and used a different notation to name what he called *analysis of gene diversity*: G_{IS} , G_{IT} , and G_{ST} . Often the two groups of indexes are used interchangeably

- ▶ This is a measure of departure from HW proportions within local demes.

$$F_{IS} = 1 - \frac{H_O}{H_S} \quad (7.1)$$

- ▶ H_O is the observed heterozygosity averaged over all demes while H_S is the expected heterozygosity averaged over all demes. F_{IS} values can range from $-1 \leq 0 \leq 1$, with positive values indicating an excess of homozygotes and negative values indicating an excess of heterozygotes (Allendorf et al., 2022).

- ▶ Often called *fixation index*, it is a measure of genetic divergence among demes.

$$F_{ST} = 1 - \frac{H_S}{H_T} \quad (7.2)$$

- ▶ H_T as the expected HW heterozygosity if the metapopulation were panmictic, and H_S is the HW proportion of heterozygotes in each separate deme and then averaged over all demes. This measure varies from 0 (demes all have equal allele frequencies) to 1 (demes are all fixed for different alleles).

- ▶ Measures the departure from HW proportions due to departures from HW proportions within local demes and divergence among demes (Allendorf et al., 2022).

$$F_{IT} = 1 - \frac{H_O}{H_T} \quad (7.3)$$

- ▶ H_O is the observed heterozygosity averaged over all demes while H_T is the expected HW heterozygosity if the metapopulation were panmictic.

- The fixation indexes described in the previous slides are related by the following expression:

$$F_{IT} = F_{IS} + F_{ST} - (F_{IS} * F_{ST}) \quad (7.4)$$

- ▶ Habitat fragmentation is a combination of two processes: reduction in total area, and the creation of separate islands
- ▶ Human induced fragmentation and habitat loss is recognized as the main cause of biodiversity loss (Baan, Alkemade, & Koellner, 2013)
- ▶ Habitat fragmentation causes an overall reduction in population size
- ▶ Habitat fragmentation also causes an overall reduction in gene flow among the patches it creates

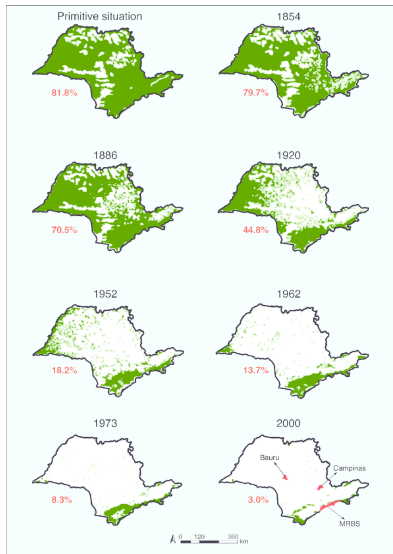


Figure 14: Fragmentation of Atlantic forest. After Nunes (2011)

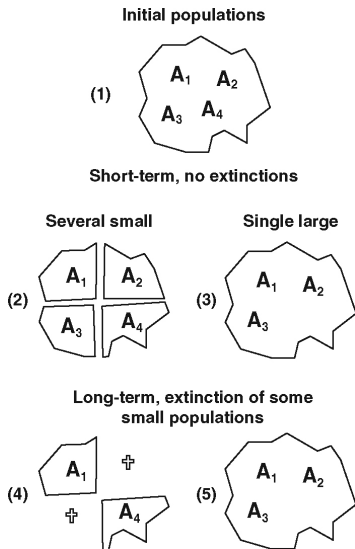


Figure 15: Single large (SL) VS several small (SS). After Frankham et al. (2010)

- ▶ Fragmentation happens in two steps:
 1. Initial genetic subdivision of the population
 2. Cumulative diversification through genetic drift, selection, inbreeding etc. etc. etc
- ▶ After an initial fragmentation, different fragments will have different initial allele frequencies. This diversification can be described by the binomial sampling variance ($\sigma_p^2 = \frac{pq}{2N}$) and the smaller the fragment, the higher the variance
- ▶ The degree of differentiation among fragments can be described by partitioning the overall inbreeding into components within and among populations (F statistics).
- ▶ We will look at some examples on estimating F statistics from genotype data.

- ▶ What is gene flow?
 - ▶ Movement of alleles between population due to migration.
- ▶ The effects of fragmentation on gene flow depend on:
 - ▶ Number of population fragments
 - ▶ Distribution of population sizes in the fragments
 - ▶ Distance between fragments
 - ▶ Dispersal ability of the species
 - ▶ Migration rates among fragments
 - ▶ Environment of the matrix among fragments
 - ▶ Times since fragmentation and extinction and recolonization rates across fragments

The black footed rock wallaby



Location	Prop. of polymorphic loci	Mean # of alleles	Average heterozygosity
Barrow Island (1)	0.1	1.2	0.05
Exmouth (2)	1.0	3.4	0.62
Wheatbelt (3)	1.0	4.4	0.56

- Data from Eldridge et al. (1999), images after Frankham et al. (2010).

Example - 1

- Populations with same allele frequencies. One with random mating the other one with inbreeding.

Popula- tion	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.4	0.2	0.4	$p = 0.5$ $q = 0.5$	0.6	0.5

- H_O observed heterozygosity averaged over all demes.
► H_S expected heterozygosity averaged over all demes.
► H_T expected heterozygosity of the metapopulation.
► $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$.

Example - 2

- Random mating populations with different allele frequencies

Popula- tion	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.04	0.32	0.64	$p = 0.2$ $q = 0.8$	0	0.32

- H_O observed heterozygosity averaged over all demes
- H_S expected heterozygosity averaged over all demes
- H_T expected heterozygosity of the metapopulation
- $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$

Example - 3

- Populations with different allele frequencies, one random mating and one with inbreeding

Popula- tion	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.14	0.13	0.74	$p = 0.2$ $q = 0.8$	0.6	0.32

- H_O observed heterozygosity averaged over all demes
- H_S expected heterozygosity averaged over all demes
- H_T expected heterozygosity of the metapopulation
- $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$

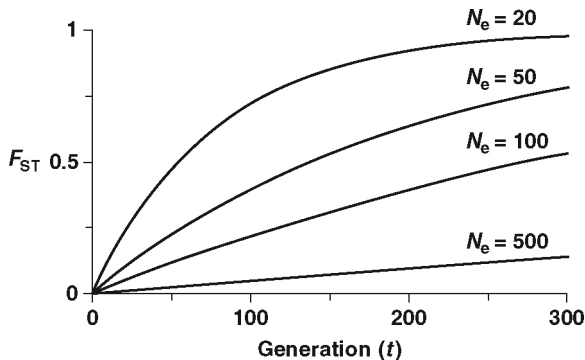
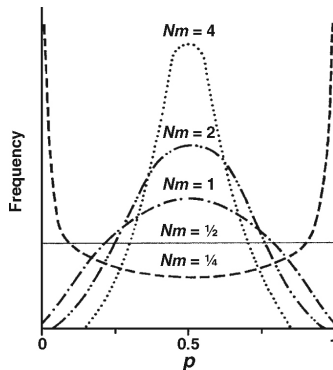


Figure 16: F_{ST} increases over generations in fragmented populations at a rate inversely dependent on population size. Values that are ≥ 0.15 indicate significant differentiation between fragments (Frankham et al., 2010).

Migration and gene flow

- ▶ Migration reduces the impact of fragmentation in a way that is dependent on the rate of gene flow
- ▶ When is gene flow sufficiently high to prevent the genetic impact of fragmentation?
- ▶ A single migrant per generation is considered sufficient to prevent differentiation between demes
- ▶ The migration rate m is the proportion of genes in a population derived from migrants per generation
- ▶ Nm is the number of migrants per generation (Frankham et al., 2010)



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